

The Supplementary Material for RecGOAT

Anonymous Author(s)

Abstract

Large models (LMs), including language, vision, and multimodal models, provide strong semantic understanding, contextual reasoning, and cross-modal fusion for multimodal recommendation. However, LM representations are inherently optimized for general semantic tasks, while recommendation models rely heavily on sparse user/item unique identity (ID) features. Existing works overlook the fundamental representational divergence between large models and recommendation systems, resulting in incompatible multimodal representations and suboptimal recommendation performance. To bridge this gap, we propose **RecGOAT**, a novel yet simple dual semantic alignment framework for LLM-enhanced multimodal recommendation, which offers theoretically guaranteed alignment capability. RecGOAT first employs graph attention networks to enrich collaborative signals by modeling item-item, user-item, and user-user relationships, leveraging user/item LM representations and interaction history. Furthermore, we design a dual-granularity progressive multimodality-ID alignment framework, which achieves instance-level and distribution-level semantic alignment via cross-modal contrastive learning (CMCL) and optimal adaptive transport (OAT), respectively. Theoretically, we demonstrate that the unified representations derived from our alignment framework exhibit superior semantic consistency and comprehensiveness. Extensive experiments on three public benchmarks show that our RecGOAT achieves state-of-the-art performance, empirically validating our theoretical insights. Our code is available at <https://anonymous.4open.science/r/RecGOAT-WSDM2027>.

CCS Concepts

• **Information systems** → **Recommender systems**; *Multimedia and multimodal retrieval*.

Keywords

Multimodal Recommendation, Large Language Models, Semantic Alignment, Graph Neural Networks, Optimal Transport

ACM Reference Format:

Anonymous Author(s). 2026. The Supplementary Material for RecGOAT. In *Proceedings of 20th ACM International Conference on Web Search and Data Mining (WSDM '27)*. ACM, New York, NY, USA, 3 pages. <https://doi.org/XXXXXX.XXXXXXX>

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WSDM '27, Hong Kong, China

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ACM ISBN 978-1-4503-XXXX-X/2018/06
<https://doi.org/XXXXXX.XXXXXXX>

In the supplementary materials, we provide additional supporting content, including: (1) a detailed case analysis of LLM-based reasoning for user profile inference in Section 3.1.3, illustrating how large language models interpret and summarize user preferences from behavioral histories; (2) complete proofs of two key lemmas and main theorem introduced in Section 3.3.2, offering further theoretical grounding for the proposed framework; and (3) runtime analysis in Section 4.6.

1 A Case Study of User Preference Reasoning

As shown in Table 1, we present the preference inference path for user 0. It can be observed that the LLM-generated user preferences accurately capture key attributes of the ground-truth target item, including product category (**Baby products**), brand (**Munchkin**), usage scenario (**home**), and visual preference (**colorful design**). Encoding such information as the user's initial textual representation effectively enhances the model's understanding of user preferences.

2 The Detailed Proofs of Lemma

The following sections provide the detailed proofs of Lemma 3.3 (Instance-level Distance Bound), Lemma 3.4 (Modality-to-Unified Error Bound) and Theorem 3.5 (Alignment Consistency and Fusion Comprehensiveness of RecGOAT) in Section 3.3.2 of the paper.

2.1 Proofs of Lemma 3.3

PROOF OF LEMMA 3.3. Let $s_i = \mathbf{z}_i^m \cdot \mathbf{z}_i$ denote the cosine similarity between the positive pair. Given the L_2 -normalization condition, the squared Euclidean distance can be strictly formulated as:

$$\|\mathbf{z}_i^m - \mathbf{z}_i\|_2^2 = \|\mathbf{z}_i^m\|_2^2 + \|\mathbf{z}_i\|_2^2 - 2\mathbf{z}_i^m \cdot \mathbf{z}_i = 2 - 2s_i. \quad (1)$$

The InfoNCE loss for a single instance i is defined as:

$$\mathcal{L}_i^m = -\log \left(\frac{\exp(s_i/\tau)}{\sum_{j=1}^B \exp(\mathbf{z}_i^m \cdot \mathbf{z}_j/\tau)} \right). \quad (2)$$

By dividing both the numerator and the denominator by the positive sample term $\exp(s_i/\tau)$, we can rewrite the loss:

$$\exp(\mathcal{L}_i^m) - 1 = \sum_{j \neq i} \exp \left(\frac{\mathbf{z}_i^m \cdot \mathbf{z}_j - s_i}{\tau} \right). \quad (3)$$

According to the Cauchy-Schwarz inequality, the cosine similarity for any negative sample $j \neq i$ satisfies $\mathbf{z}_i^m \cdot \mathbf{z}_j \geq -1$. Since the exponential function is strictly monotonically increasing, we can establish a strict lower bound for the summation over the $B - 1$ negative samples:

$$\sum_{j \neq i} \exp \left(\frac{\mathbf{z}_i^m \cdot \mathbf{z}_j - s_i}{\tau} \right) \geq (B - 1) \exp \left(\frac{-1 - s_i}{\tau} \right). \quad (4)$$

Combining Eq. (3) and Eq. (4), we have:

$$(B - 1) \exp \left(\frac{-1 - s_i}{\tau} \right) \leq \exp(\mathcal{L}_i^m) - 1. \quad (5)$$

Table 1: Case Study of LLM-based Preference Reasoning on the *Baby* Dataset

Input (Prompt & User Context)	Output (LLM Reasoning & Real Target Item)
User ID: 0 Prompt: You are a professional data analyst. Your task is to analyze a user's interaction history to infer their preferences. User Information • User ID: {user id} • Interaction History: {item ids & text description} Please conduct a structured reasoning by two steps: 1. Identify Common Attributes Across Items - Extract shared characteristics from perspectives such as category, function, style, material, usage scenario, and price range. Organize the findings in a list format. 2. Summarize Preferences Across Multiple Dimensions - Systematically summarize the user's preferences based on the following dimensions, with a brief explanation of supporting evidence: • Style & Aesthetics: e.g., design style, color preference, visual element tendencies • Function & Usability: e.g., functional needs, usage scenarios, interaction experience • Context & Emotion: e.g., emotional value associated with items, social attributes, life scenarios • Other Dimensions: e.g., brand loyalty, price sensitivity, attention to sustainability Output Format: <think> reasoning process here </think> <answer> answer here </answer> Please provide your analysis below:	LLM Response: <think>... ..</think> <answer> 1. Identify Common Attributes Across Items - Category: All items fall under Baby products, ... - Brand: Repeated engagement with trusted brands like Philips Avent, Munchkin, Britax, and Petunia Pickle Bottom. - Usage Scenario: Items cater to home, travel, and on-the-go use (e.g., compact breast pump, travel bag). 2. Summarize Preferences Across Multiple Dimensions Style & Aesthetics - Design Preference: Combines functional simplicity with playful or trendy visuals (e.g., floral-patterned bottles, colorful silicone spoons). - Color Tendencies: Neutral and pastel hues (pink, green) alongside vibrant accents (red stroller). - Brand Aesthetics: Appreciation for stylish yet practical brands like Petunia Pickle Bottom (cosmopolitan charm) and Philips Avent ... - Brand Loyalty: Clear preference for reputable brands (Philips Avent appears twice; Britax and Munchkin multiple times). ... Key Takeaway: The user prioritizes safe, functional, and stylish baby products from trusted brands, balancing practicality with aesthetic appeal for both home and travel scenarios. </answer> ----- Target Item for Next-Buy Prediction (Ground Truth): "B000GK7AMU": Munchkin Five Sea Squirts. Brand: Munchkin. Category: Baby. Description: Fill them up and squirt them out. They make bath time a real blast for babies. The lovable and colorful sea characters provide creative play and are an amusing way for moms to rinse away soap. Just add them to your water and expect an ocean of fun.

Solving this inequality for s_i yields the strict lower bound for the positive sample similarity:

$$s_i \geq -1 - \tau \ln \left(\frac{\exp(\mathcal{L}_i^m) - 1}{B - 1} \right). \quad (6)$$

Substituting Eq. (6) into Eq. (1), we obtain the strict upper bound for the squared instance distance:

$$\|z_i^m - z_i\|_2^2 \leq 4 + 2\tau \ln \left(\frac{\exp(\mathcal{L}_i^m) - 1}{B - 1} \right). \quad (7)$$

When the model converges, the InfoNCE loss is sufficiently small, which satisfies the inequality $\exp(x) - 1 \leq 2x$. Applying this to Eq. (7), taking the expectation over all instances, and utilizing Jensen's inequality (since $\ln(\cdot)$ is a concave function), we derive:

$$\mathbb{E}_i [\|z_i^m - z_i\|_2^2] \leq 4 + 2\tau \ln \left(\frac{2\mathcal{L}_{CMCL}}{B - 1} \right), \quad (8)$$

where $\mathcal{L}_{CMCL} = \mathbb{E}_i [\mathcal{L}_i^m]$. Finally, by applying the Cauchy-Schwarz inequality $(\mathbb{E}_i [\|z_i^m - z_i\|_2])^2 \leq \mathbb{E}_i [\|z_i^m - z_i\|_2^2]$, we obtain the expected distance bound:

$$\mathbb{E}_i [\|z_i^m - z_i\|_2] \leq \sqrt{4 + 2\tau \ln \left(\frac{2\mathcal{L}_{CMCL}}{B - 1} \right)}. \quad (9)$$

This completes the proof. $\square$

2.2 Proofs of Lemma 3.4

PROOF OF LEMMA 3.4. From the definitions of $\epsilon_m(f)$ and $\epsilon_F(f)$, and applying the triangle inequality, we have for any fixed user $\mathbf{u}$ and a item i :

$$|f(\mathbf{u}, z_i^m) - f^*(\mathbf{u}, v_i)| - |f(\mathbf{u}, z_i) - f^*(\mathbf{u}, v_i)| \leq |f(\mathbf{u}, z_i^m) - f(\mathbf{u}, z_i)|. \quad (10)$$

Taking the expectation with respect to the unified (target) distribution Q yields and applying the Jensen's inequality:

$$\left| \mathbb{E}_{z_i \sim Q} [|f(\mathbf{u}, z_i^m) - f^*(\mathbf{u}, v_i)|] - \epsilon_F(f) \right| \leq \mathbb{E}_{z_i \sim Q} [|f(\mathbf{u}, z_i^m) - f(\mathbf{u}, z_i)|]. \quad (11)$$

Let the error function for a given user u and representation z be defined as $g(z) = |f(u, z) - f^*(u, z)|$. For any representations z_1 and z_2 , applying the reverse triangle inequality yields:

$$|g(z_1) - g(z_2)| = \left| |f(u, z_1) - f^*(u, z_1)| - |f(u, z_2) - f^*(u, z_2)| \right| \leq |f(u, z_1) - f(u, z_2)| + |f^*(u, z_1) - f^*(u, z_2)|. \quad (12)$$

Given that f is K -Lipschitz (Assumption 3.1) and f^* is L^* -Lipschitz with respect to z (Assumption 3.2), we have:

$$|g(z_1) - g(z_2)| \leq K\|z_1 - z_2\| + L^*\|z_1 - z_2\| = (K + L^*)\|z_1 - z_2\|. \quad (13)$$

Thus, $g(z)$ is indeed $(K + L^*)$ -Lipschitz continuous.

By the Kantorovich-Rubinstein duality of the 1-Wasserstein distance, for the $(K + L^*)$ -Lipschitz function g , we have:

$$|\epsilon_m(f) - \mathbb{E}_{z_i \sim Q}[|f(u, z_i^m) - f^*(u, v_i)|]| \leq (K + L^*) \cdot \mathcal{W}_1(P^m, Q). \quad (14)$$

Finally, combining Eq. (11) and Eq. (14) via the triangle inequality gives:

$$|\epsilon_m(f) - \epsilon_F(f)| \leq (K + L^*) \cdot \mathcal{W}_1(P^m, Q) + \mathbb{E}_{z_i \sim Q}[|f(u, z_i^m) - f(u, z_i)|]. \quad (15)$$

Using the K -Lipschitz continuity of f again, the second term is bounded by $K \cdot \mathbb{E}_i\|z_i^m - z_i\|$, yields the final result:

$$|\epsilon_m(f) - \epsilon_F(f)| \leq (K + L^*) \cdot \mathcal{W}_1(P^m, Q) + K \cdot \mathbb{E}_i\|z_i^m - z_i\|. \quad (16)$$

This completes the proof. $\square$

2.3 Proofs of Theorem 3.5

PROOF OF THEOREM 3.5. Starting from Lemma 3.4, we have for any modality m :

$$\epsilon_m(f) - \epsilon_F(f) \leq (K + L^*) \cdot \mathcal{W}_1(P^m, Q^{id}) + K \cdot \mathbb{E}_i\|z_i^m - z_i\|. \quad (17)$$

Applying the bound from Lemma 3.3 to instance-level term yields:

$$\epsilon_m(f) \leq \epsilon_F(f) + (K + L^*) \cdot \mathcal{W}_1(P^m, Q^{id}) + K \cdot \sqrt{4 + 2\tau \ln \left(\frac{2\mathcal{L}_{CMCL}}{B - 1} \right)}. \quad (18)$$

Since this inequality holds for all $m \in \mathcal{M}$, taking the maximum over modalities on the left side yields the **Consistency Guarantee**:

$$\max_{m \in \mathcal{M}} \epsilon_m(u) \leq \epsilon_F(u) + (K + L^*) \cdot \mathcal{W}_1(P^m, Q^{id}) + \sqrt{4K^2 + 2\tau K^2 \ln \left(\frac{2\mathcal{L}_{CMCL}}{B - 1} \right)}.$$

Similarly, from Lemma 3.4 we also have:

$$\epsilon_F(f) - \epsilon_m(f) \leq (K + L^*) \cdot \mathcal{W}_1(P^m, Q^{id}) + K \cdot \mathbb{E}_i\|z_i^m - z_i\|. \quad (19)$$

Applying the same substitution from Lemma 3.3 gives:

$$\epsilon_F(u) \leq \epsilon_m(u) + (K + L^*) \cdot \mathcal{W}_1(P^m, Q^{id}) + K \cdot \sqrt{4 + 2\tau \ln \left(\frac{2\mathcal{L}_{CMCL}}{B - 1} \right)}. \quad (20)$$

As this is valid for all $m \in \mathcal{M}$, the tightest bound is achieved by taking the minimum over modalities on the right-hand side, resulting in the **Comprehensiveness Guarantee**:

$$\epsilon_F(u) \leq \min_{m \in \mathcal{M}} \left\{ \epsilon_m(u) + (K + L^*) \cdot \mathcal{W}_1(P^m, Q^{id}) + \sqrt{4K^2 + 2\tau K^2 \ln \left(\frac{2\mathcal{L}_{CMCL}}{B - 1} \right)} \right\}. \quad (21)$$

$\square$

Table 2: Comparison of training and inference time on the Baby Dataset.

Model	LightGCN	FindRec	IRLLRec	RecGOAT
Training Time (s/training_epoch)	1.33	2.95	3.41	3.12
Inference Time (s/evaluation_set)	0.98	1.23	2.03	1.37

3 Runtime Analysis (RQ5)

To evaluate computational efficiency, we report the training and inference times of RecGOAT alongside three representative baselines on the Baby dataset, as summarized in Table 2. Specifically, we select one state-of-the-art and recent model from each category of recommendation paradigms: LightGCN (Traditional ID-based Methods), FindRec (Multimodal Methods), and IRLLRec (LM-enhanced Methods). All models are evaluated under identical hyperparameters and computational resources for a fair comparison. As expected, LightGCN achieves the fastest training and inference speeds owing to its highly simplified linear graph convolutional architecture. Compared to more complex baselines, our proposed RecGOAT outperforms IRLLRec in both training time (3.12 s/epoch) and inference time (1.37 s/evaluation set). Although RecGOAT incurs slightly higher time costs than FindRec, the overall computational overhead remains within the same order of magnitude and is highly acceptable. This indicates that while introducing superior recommendation capabilities, RecGOAT maintains highly competitive computational efficiency, demonstrating its strong potential for deployment in real-world recommender systems.